

Real-Time Collaborative Kanban Board Assignment (Multi-Organization + Multi-Tenant)

Problem Statement

Build a real-time collaborative task board where multiple users across organizations and their branches (tenants) can collaborate with strict data isolation and live synchronization.

Core Requirements

Backend

- 1 CRUD APIs for Organizations, Tenants (Branches), Boards, Columns, Tasks
- 2 Multi-organization → multi-tenant hierarchy
- 3 Strict data isolation at tenant level
- 4 Role-based access control (Org Admin, Tenant Admin, Member)
- 5 Real-time sync using WebSockets or SSE
- 6 Concurrency handling to avoid conflicting updates

Database Design

- 1 Tables: organizations, tenants, users, memberships, boards, columns, tasks, activity_logs
- 2 Hierarchy: Organization → Tenants → Boards
- 3 All business data scoped by tenant_id
- 4 Efficient ordering of tasks within columns
- 5 Versioning or timestamps for conflict handling

Frontend

- 1 Organization selector + Tenant (branch) switcher
- 2 Kanban board UI with drag-and-drop
- 3 Task detail panel
- 4 Real-time updates across users in same tenant

Advanced Requirements

- 1 Presence indicators per tenant

- 2 Scoped WebSocket channels per tenant
- 3 Optimistic UI updates
- 4 Undo/redo
- 5 Activity logs per tenant
- 6 Conflict resolution strategy

Edge Cases

- 1 Cross-tenant data leakage prevention
- 2 Cross-organization isolation
- 3 Concurrent edits
- 4 Drag-drop conflicts
- 5 Network recovery

Deliverables

- 1 Backend + real-time server
- 2 Frontend UI
- 3 Database schema (ER diagram)
- 4 README (architecture, tenancy model, trade-offs)
- 5 API + WebSocket documentation

Evaluation Criteria

- 1 Architecture (30%) – modular + tenancy modeling
- 2 Correctness (20%) – edge cases
- 3 Database Design (15%) – tenant isolation
- 4 Frontend Quality (15%)
- 5 Advanced Thinking (20%) – concurrency + scalability